Can Ambient WiFi Support Vehicular Multipath SLAM? Virtual-Anchor Localization, Commodity Front-End Limits, and a Delay-Convention Pitfall in Ray-Traced Evaluation

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Abstract

Ambient WiFi is radiated by existing infrastructure and received by hardware a vehicle already carries, which makes it an attractive low-cost aid for vehicular simultaneous localization and mapping (SLAM). We ask what commodity channel state information (CSI) can actually contribute, using ray-traced channels (Sionna RT) in three outdoor scenarios, wheel-grade odometry as the baseline that any aid must beat, and a virtual-anchor FastSLAM backend in which each wall reflection is represented by the access point mirrored in that wall. With exact multipath parameters the geometry works: WiFi reduces odometry drift by 66–79 % and, beside a single reflecting wall, outperforms a simulated spinning LiDAR. With a commodity front-end it does not: joint delay–angle MUSIC on a four-element linear array returns only about 15 % facade reflections, about a third of detections match no propagation path, and a further 19–34 % are real paths reported at their front/back mirror angle. These errors persist across frames, are confirmed as false anchors, and make WiFi-aided SLAM worse than dead reckoning in every scenario; the share of unexplained detections does not fall with bandwidth from 20 to 160 MHz. We also report a delay-convention pitfall in ray-traced CSI studies: channel frequency responses whose delays are referenced to the first arrival, as on real hardware, were read as absolute path lengths in our earlier analyses. This produced a spurious several-metre range bias and an inflated

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phantom rate, both of which vanish once excess delay is modelled explicitly. Code, cached channels, and firmware are released openly.

Keywords: WiFi sensing, channel state information, multipath-assisted SLAM, virtual anchors, vehicular localization, ray tracing

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1. Introduction

Exteroceptive sensors are the expensive part of a vehicle's localization and mapping stack. WiFi is the opposite: access points (APs) already illuminate most built environments, and commodity receivers that report channel state information (CSI) cost a few dollars. Multipath components in that CSI carry geometric information, since each specular reflection arrives along a path whose length and angle are fixed by the reflector and the transmitter [1, 2]. Whether a moving vehicle can turn that information into better localization, or into a map, depends on two separate questions: is the geometry informative, and can a commodity front-end extract it?

This paper separates those questions. We evaluate ambient-WiFi SLAM in ray-traced outdoor scenes against the baseline that matters for a vehicle, wheel-grade odometry, and against a simulated LiDAR given the same odometry. The back-end is a virtual-anchor FastSLAM [3]: each wall reflection is represented by the AP mirrored in that wall, a point that is fixed in the world and whose range equals the bistatic path length. We feed it either exact multipath parameters (an oracle) or the output of joint delay–angle MUSIC [4, 5] applied to noisy CSI from a four-element array.

The work also corrects our own earlier analyses [6], which reported centimetre-level WiFi localization at parity with LiDAR and attributed a mapping failure to a "phantom ceiling" with a several-metre range bias. Re-verification found three defects: the estimator was driven by ground-truth odometry, its measurement update carried no position information, and CSI delays referenced to the first arrival were read as absolute path lengths. Section 5 documents the last of these, because it is easy to repeat in any study that synthesizes CSI from a ray tracer.

Contributions.

- A ray-traced evaluation of ambient-WiFi vehicular SLAM with a realistic odometry baseline, paired LiDAR runs, a straight and a lane-change trajectory, and ten seeds (Sections 3–4).

- Evidence that the virtual-anchor geometry is informative: exact multipath parameters reduce odometry drift by 66–79 % (Section 4.1).
- A decomposition of what a commodity front-end returns, showing that unexplained detections and single-array front/back folding, not range resolution, prevent that gain from being realized (Sections 4.2–4.4).
- A description of the excess-delay pitfall, its quantitative effect, and a measurement model that handles it (Section 5).
- Hardware evidence from two ESP32-S3 boards that commodity CSI indeed fixes delays only relative to the first arrival (Section 6), and an open artifact.

2. Related work

WiFi and CSI sensing is mature for indoor, largely static settings: through-wall and device-free sensing [7, 8, 9], CSI ranging and imaging [10, 11, 12], and pose recovery [13, 14]. Decimetre AP-side localization with super-resolution has been shown with commodity cards [15], and the lack of absolute time of flight in commodity CSI is well documented [16]; it is the reason the delay convention in Section 5 matters.

Multipath-assisted localization and SLAM treats specular reflections as virtual anchors whose positions are learned jointly with the agent's [1, 2, 17, 18, 19]. These systems typically use wideband, synchronized, or carefully calibrated signals. WiFi-based SLAM otherwise localizes against known APs or radio fingerprints [20, 21], or uses WiFi to assist a primary sensor [22, 23, 24].

Integrated sensing and communication envisions shared waveforms for both functions [25, 26, 27, 28]; 60 GHz 802.11ad has been used as a radar [29, 30], and a passive radar on a moving platform has been demonstrated with 5G [31]. Our setting, ambient sub-7 GHz WiFi received on a moving vehicle outdoors with commodity array sizes, sits between these lines; the question we answer is how much of the multipath-SLAM machinery survives a commodity front-end.

3. System model and method

3.1. Bistatic geometry and virtual anchors

A vehicle at horizontal position $\mathbf{p}$ receives signals from N_{AP} ambient APs at known positions $\mathbf{a}_i$. A single specular reflection from a planar facade follows the path $\mathbf{a}_i \to \mathbf{r} \to \mathbf{p}$; mirroring the AP in the facade gives the virtual anchor (VA) $\mathbf{v}$, and the reflection is indistinguishable from a line-of-sight (LOS) signal from $\mathbf{v}$:

$$\rho = \|\mathbf{v} - \mathbf{p}\|, \qquad \phi = \mathrm{atan2}(\mathbf{v} - \mathbf{p}), \tag{1}$$

where ρ is the (horizontal) bistatic path length and ϕ the arrival azimuth. Because $\mathbf{v}$ is fixed in the world, repeated observations constrain $\mathbf{p}$. The reflection point $\mathbf{r}$ follows from (ρ, ϕ) and $\mathbf{a}_i$ by the bistatic ellipse solve and is what we map. All lengths are projected to the ground plane using the known AP and receiver heights, which a facade bounce preserves.

3.2. Excess-delay measurement model

Commodity CSI does not provide absolute time of flight; delays are known only relative to the first arrival [16]. With an LOS first arrival the measured excess length is

$$e = \rho_{\mathrm{3D}} - \|\mathbf{a}_i - \mathbf{p}\|_{\mathrm{3D}} \quad \Longrightarrow \quad \rho_{\mathrm{3D}}(\mathbf{p}) = e + \|\mathbf{a}_i - \mathbf{p}\|_{\mathrm{3D}}, \tag{2}$$

so the path length used in (1) depends on the hypothesized pose and the AP position. The filter evaluates (2) per particle. This is the model whose omission is discussed in Section 5.

3.3. Odometry

Wheel odometry is modelled with a per-run scale error ($\sigma = 1\,\%$), an initial heading error ($\sigma = 0.5^\circ$), a heading random walk ($0.2^\circ/\sqrt{\mathrm{s}}$), and white velocity noise ($0.05\,\mathrm{m/s}$). Dead reckoning on this odometry is reported beside every result, and the WiFi and LiDAR runs of a given seed use the identical odometry draw.

3.4. Virtual-anchor FastSLAM

Each of 200 particles carries a pose and an extended Kalman filter per VA [3]. A detection is associated, per particle, with the VA of the same AP that minimizes the Mahalanobis distance of (ρ, ϕ), gated at the 99 % χ^2_2 level; associated detections update the VA and weight the particle by the innovation likelihood, and unexplained detections initialize a tentative VA. A VA is confirmed after five associations, and a tentative VA unseen for ten frames is deleted. The map consists of reflection points of detections associated with confirmed VAs, located from the posterior pose. Measurement standard deviations are fixed a priori: 0.3 m and 2° for the oracle, $0.5\,c/B$ and 5° for MUSIC at bandwidth B. A single uniform linear array (ULA) reports only the forward azimuth branch; an optional *two-branch* model scores each detection against both ϕ and $\pi - \phi$ and seeds a tentative VA for each.

3.5. Channels, scenes, and front-ends

Channels are ray traced with Sionna RT 2.0 [32] at 5.2 GHz (maximum depth three) and stored once per scene, including the noise-free channel frequency response at 20, 40, 80, and 160 MHz and the full table of propagation paths. Seeds vary the receiver noise (20 dB SNR) and the odometry. The receiver is a four-element half-wavelength ULA with 128 subcarriers. Three scenarios are used: a *controlled* metal wall (30 m, 120 frames, three APs), a *street canyon* with metal facades (60 m, 240 frames, three APs), and the same canyon driven on a lane-change S-curve of 2.5 m amplitude. The *oracle* front-end uses the true single-scatter, non-floor paths in the excess-delay convention; the *realistic* front-end is joint 2-D MUSIC with 2-D spatial smoothing [4, 5] estimating three paths per AP.

3.6. Baseline and metrics

The LiDAR baseline is an Ouster-class scanner ray-cast against the scene geometry with a scan-to-map ICP back-end, externally validated on KITTI sequence 04 (1.16 m absolute trajectory error over 394 m) [33], and seeded with the same noisy odometry. We report the absolute trajectory error (ATE), occupancy intersection-over-union (IoU, 1 m cells) against facade footprints, and map accuracy. To characterize the front-end, every detection is matched per frame to the true paths of the same AP using the normalized cost $\sqrt{(\Delta\rho/3\,\mathrm{m})^2 + (\Delta\phi/10°)^2} \leq 3$; a detection that matches only after mirroring the true azimuth onto the forward branch is counted as *folded*, and one that matches neither is *unexplained*.

Table 1: Absolute trajectory error (m) under wheel-grade odometry, ten seeds.

	Controlled wall	Street canyon	Street, S-curve
Dead reckoning	0.211 ± 0.109	0.470 ± 0.260	0.472 ± 0.259
WiFi, oracle	$\mathbf{0.072 \pm 0.026}$	0.101 ± 0.045	0.100 ± 0.047
WiFi, MUSIC	0.340 ± 0.123	1.202 ± 0.241	1.065 ± 0.199
WiFi, MUSIC, two-branch	0.390 ± 0.110	0.527 ± 0.167	0.488 ± 0.266
LiDAR (ICP)	0.172 ± 0.117	$\mathbf{0.027 \pm 0.001}$	$\mathbf{0.028 \pm 0.000}$

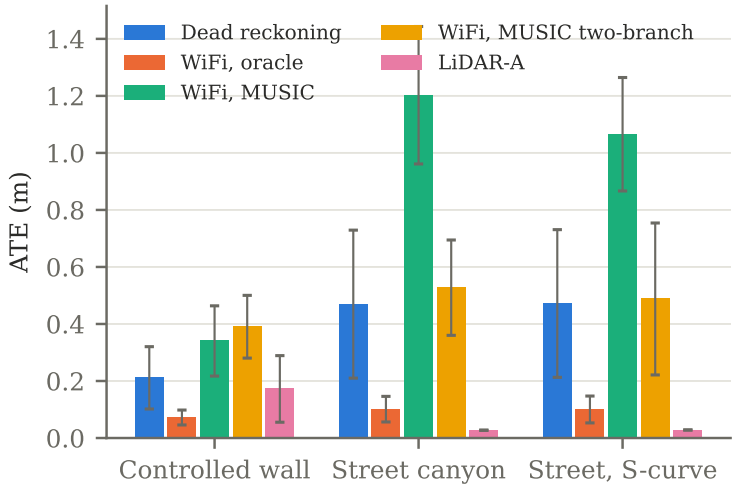


Figure 1: Trajectory error under the same wheel-grade odometry. Exact multipath parameters make WiFi a useful aid; joint MUSIC on a four-element array makes it harmful.

4. Results

Results are mean ± standard deviation over ten seeds at 160 MHz unless stated otherwise.

4.1. Localization

Table 1 and Fig. 1 give the central result. With exact multipath parameters, WiFi reduces the odometry error by 66 % beside the single wall and by 79 % in both street runs; the filter confirms exactly the three and seven VAs present. Beside a single wall WiFi also outperforms LiDAR, whose scan matching cannot constrain motion along a featureless plane, whereas the VAs

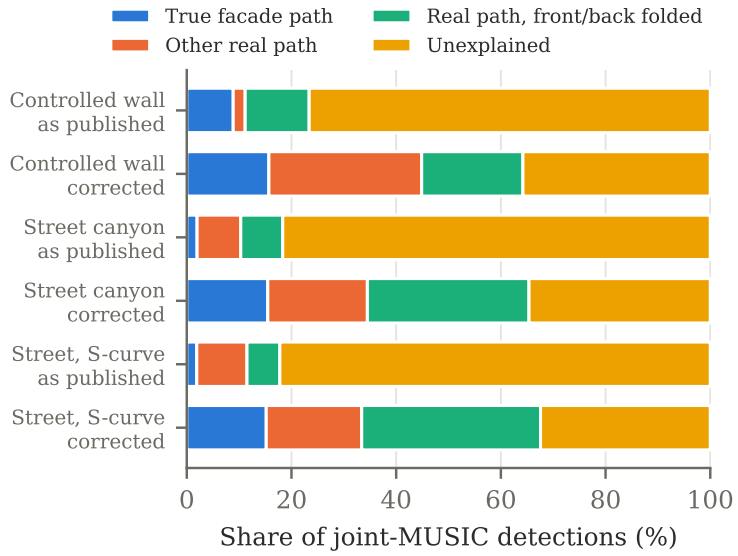


Figure 2: Composition of joint-MUSIC detections, as published in our earlier analysis (excess delay read as absolute length) and corrected. Only about 15 % of detections are facade reflections in every scenario.

of three APs observe the wall from different directions. In the street canyon LiDAR is four times better again.

With the realistic front-end the same filter is worse than dead reckoning in every scenario. It confirms 28 VAs beside the wall and 101–116 in the canyon, against three and seven real ones: erroneous detections recur from frame to frame often enough to pass the five-observation promotion test, and once confirmed they pull the particles. The two-branch model, which accounts for the array's front/back ambiguity, removes most of the excess error in the canyon but only reaches dead-reckoning parity. The trajectory shape makes no material difference.

4.2. What the commodity front-end returns

Table 2 and Fig. 2 explain the localization failure. Only one detection in six is a facade reflection, the only kind that defines a VA. A fifth to a third are genuine paths reported at their mirror angle, which a single ULA cannot resolve; these are not estimator artefacts, and a second array axis or a vehicle-heading change would disambiguate them. About a third match no

Table 2: Composition of joint-MUSIC detections (%) and median signed range error of matched facade detections, at 160 MHz.

	Controlled	Street	S-curve
True facade path	15.7	15.5	15.2
Other real path (LOS, floor, multi-bounce)	29.2	19.0	18.2
Real path, front/back folded	19.3	30.8	34.1
Unexplained	35.8	34.6	32.4
Facade range error, corrected (m)	+0.03	−0.06	−0.06
As published: unmatched share	88.8	89.7	88.5
As published: facade range error (m)	−6.35	−2.19	−6.62

Table 3: Joint-MUSIC detection composition and ground-truth-pose map IoU versus bandwidth (five seeds).

	Controlled		Street	
Bandwidth	Unexplained (%)	IoU	Unexplained (%)	IoU
20 MHz	44.4	0.474	37.7	0.012
40 MHz	36.9	0.341	43.3	0.020
80 MHz	35.9	0.331	33.3	0.027
160 MHz	35.8	0.349	34.6	0.031

propagation path within the generous tolerance. The facade detections that are found are accurate in range (median error below 0.1 m), so the problem is not bias or precision but the composition of the detection set.

4.3. Mapping under ground-truth poses

Scoring the map at ground-truth poses isolates the front-end from localization. Beside the wall the oracle reaches an IoU of 0.767 and MUSIC 0.349 ± 0.017 (Fig. 3); the as-published reading scored 0.000. In the canyon the oracle itself reaches only 0.075, because three APs illuminate a small fraction of the facade footprints, and MUSIC reaches 0.03. Mapping of the illuminated facades is therefore possible with commodity CSI in a clean geometry, but it carries clutter from the misplaced detections and is bounded by illumination coverage.

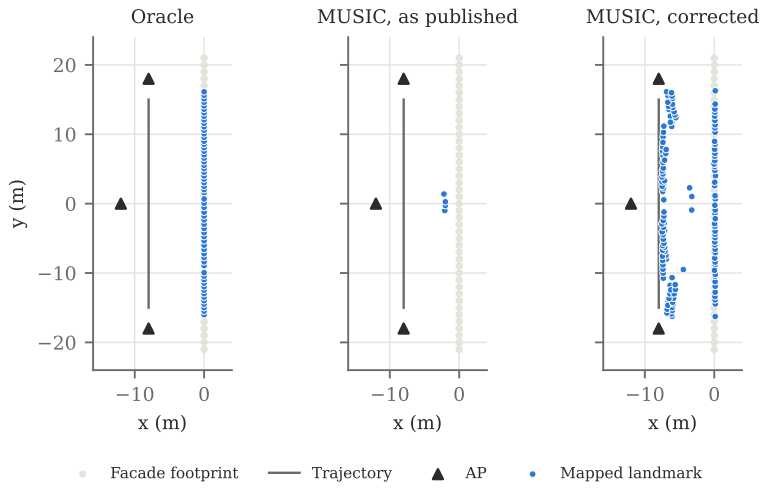


Figure 3: Maps of the controlled wall triangulated at ground-truth poses (one seed). The as-published reading of the realistic detections places almost nothing on the wall; with the excess-delay model the wall is recovered, together with clutter near the trajectory.

4.4. Bandwidth

Eightfold more bandwidth does not reduce the unexplained share (Table 3). The limit is set by the four-element aperture and by model mismatch in a coherent, single-snapshot multipath channel, not by range resolution.

4.5. Robustness to AP-position error and measurement loss

Unlike in our earlier analysis, AP-position error now enters the measurement through (2), and measurement loss now matters because the odometry drifts. Fig. 4 reports five seeds per point; dead reckoning is 0.158 m (controlled) and 0.370 m (street) for these seeds. **AP-position error is critical.** With exact multipath parameters the oracle's advantage over odometry disappears once the AP coordinates are wrong by about 0.5 m beside the wall (0.242 m against 0.158 m) and by about 1 m in the canyon (0.479 m against 0.370 m); at 8 m of error the oracle is two to three times worse than dead reckoning. Every excess-delay measurement inherits the AP error through (2), so an ambient-WiFi aid requires AP positions surveyed to sub-metre accuracy, or estimated jointly. **Measurement loss is tolerated.** A total blackout of 40 frames (17 % of the street run, 33 % beside the wall) raises the oracle's error only from 0.083 to 0.106 m in the canyon and from 0.057 to 0.074 m beside the wall; the filter falls back on odometry and re-acquires

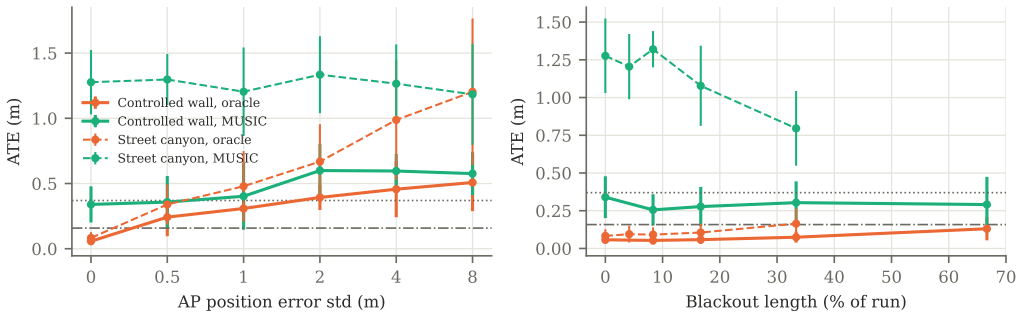


Figure 4: Sensitivity to AP-position error (left) and to a total measurement blackout starting at one third of the run (right); solid: controlled wall, dashed: street canyon; horizontal lines: dead reckoning. Five seeds.

the confirmed VAs. For MUSIC, removing measurements makes the error *smaller* (1.277 to 0.796 m for an 80-frame blackout in the canyon), consistent with its detections doing harm.

4.6. Geometry, carrier, and bandwidth: the radar ablation

[Re-run of the CFAR ablation (cells A–D) and of the MUSIC row with the excess-delay correction (M*) from results/ablation_*.json.]

5. The excess-delay pitfall

Sionna RT's channel-frequency-response routine references all delays to the first arriving path by default. That is realistic, because commodity CSI has no absolute time of flight, but a back-end that treats the resulting delays as absolute bistatic path lengths under-ranges every detection by the LOS distance $\|\mathbf{a}_i - \mathbf{p}\|$. In our earlier analyses this produced three findings that were artefacts: a median range bias of several metres (6.35 m beside the wall; 0.03 m once corrected), an unmatched share near 89 % (55–67 % corrected, of which a third to a half is front/back folding), and a map IoU of zero at ground-truth poses (0.349 beside the wall corrected). It also explains why the earlier mapping floor did not move with bandwidth or aperture: no resolution can remove an offset equal to the LOS distance. The radar chain in the same code read absolute delays directly and was unaffected.

Two safeguards would have caught it. First, triangulating the *true* path parameters and the *estimated* ones at ground-truth poses and comparing the

two isolates any systematic offset in the estimator chain. Second, a synthetic single-path test of the full chain, from CSI synthesis through detection, exposes a reference-convention mismatch immediately. We recommend both for any study that synthesizes CSI from a ray tracer.

6. Hardware: what commodity CSI provides

Two ESP32-S3 boards [34, 35], one transmitting HT40 frames and one logging CSI, were used to confirm the measurement convention on real silicon. From a static capture we parse valid 40 MHz high-throughput records whose long training field carries 128 subcarriers with 114 non-null, matching the physical active mask. The delay profile obtained by coherent averaging places the LOS tap at zero delay with no absolute reference, which is the excess-delay convention of (2). This bench validates the CSI reading and the convention, not a SLAM system; measuring detection composition against surveyed reflectors on this hardware requires a large surveyed space and is left as future work. The firmware and the offline chain are released so that the measurement can be reproduced for the cost of two development boards.

7. Discussion

What would close the gap. The oracle results show that the geometry is informative enough to matter. The realistic results identify what stands in the way, in order of share: an unexplained third of the detections, front/back folding, and non-facade real paths. Folding is a property of the single linear array and would be removed by a second array axis or by exploiting heading changes; the unexplained detections call for front-ends with explicit model-order selection or sparse estimation, and for VA promotion rules that test geometric consistency rather than mere persistence.

Limitations. All channels are simulated; the hardware result confirms the measurement convention, not the detection statistics. The scenes use perfectly conducting facades, which favour specular reflection. Three APs is a modest density, which limits map coverage in the canyon. The LOS-first assumption in (2) fails where the direct path is blocked. Heading is assumed known for converting array angles to world azimuth.

Correction of earlier claims. Our earlier unpublished manuscripts reported WiFi localization at parity with LiDAR, robustness to AP-position error and to measurement blackouts, a fusion design rule, and cost-normalized

accuracy. Those results relied on ground-truth odometry and a measurement update without position information and are withdrawn; the mapping-mechanism results that relied on the delay convention are corrected in Section 5.

8. Conclusion

Ambient WiFi received on a vehicle carries enough multipath geometry to reduce wheel-odometry drift substantially when reflections are estimated exactly, and a virtual-anchor FastSLAM uses it without knowing the facades in advance. A commodity four-element array with joint MUSIC does not deliver that geometry: most detections are not facade reflections, a large share are mirrored by the array's front/back ambiguity, and the errors persist long enough to be confirmed as anchors, so WiFi-aided SLAM ends up worse than odometry alone. Bandwidth does not change this. Mapping of illuminated facades is possible at known poses. Finally, ray-traced CSI studies must model delays relative to the first arrival explicitly; reading them as absolute path lengths manufactures a range bias and a phantom rate that are not properties of the sensor.

Data and code availability

All code, configurations, cached ray-traced channels (regenerable), result files, and ESP32 firmware are available under AGPL-3.0 at `https://github.com/mulhamfetna/wifi-radar-slam`.

Declaration of competing interest

The author declares no competing interests.

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